



NETSEC HUB

Professional Networking & Cybersecurity Training

Palo Alto Networks Firewall– Enterprise Firewall Administration & Security

- ***Mode: Online***
- ***Duration: 2.5-3 Months***

Course Overview:

Palo Alto Firewall Industry Skill Development Program:

This course is designed to develop practical firewall administration and network security skills required for entry-level Firewall Engineer and Network Security Engineer roles. The focus of this program is not limited to certification exam preparation; instead, it focuses on understanding next-generation firewall concepts, security policy implementation, traffic processing, troubleshooting, and enterprise firewall deployment approaches.

Students will learn how modern enterprise firewalls are designed, configured, and managed using industry-standard security practices. The course includes live demonstrations, practical configuration sessions, troubleshooting scenarios, Q&A discussions, and interview preparation to help students develop job-oriented firewall skills.

Hands-on concepts will be demonstrated using firewall simulation environments such as Palo Alto VM-Series in EVE-NG. Students will learn practical implementation of security policies, NAT, VPN, traffic inspection, and firewall administration according to the course topics. *(Lab files and personal lab infrastructure are not included with the course.)*

Course Modules:

Module 1: Firewall Fundamentals & Palo Alto Introduction

- Introduction to Network Security and Firewalls
 - Traditional Firewall vs Next-Generation Firewall (NGFW)
 - Palo Alto Firewall Architecture and PAN-OS Overview
 - Firewall Deployment Modes (Layer 2, Layer 3 and Virtual Wire)
 - Management Interface, Initial Setup and Device Configuration
 - CLI and Web Interface Navigation
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Module 2: Interfaces, Zones & Security Policies

- Interface Configuration and Interface Types
 - Security Zones and Zone-Based Security
 - Virtual Router and Basic Routing Concepts
 - Security Policy Fundamentals and Rule Processing
 - Security Policies Based on Applications, Users, Addresses and Services
 - Security Policy Management and Best Practices
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Module 3: Network Address Translation (NAT)

- NAT Fundamentals and Working
 - Source NAT (SNAT), Destination NAT (DNAT) and Static NAT
 - Dynamic IP and Port NAT (PAT)
 - NAT Policy Configuration and Rule Processing
 - NAT Troubleshooting and Verification
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Module 4: Routing & Network Services

- Static Routing and Default Route Configuration
 - Dynamic Routing Overview (OSPF and BGP Integration)
 - Virtual Router Configuration
 - DHCP Server and DHCP Relay Configuration
 - Network Service Integration and Troubleshooting
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Module 5: Security Features & Traffic Inspection

- Application Identification (App-ID), User Identification (User-ID) and Content Inspection (Content-ID)
 - Security Profiles and Threat Prevention
 - Antivirus, Anti-Spyware and Vulnerability Protection
 - URL Filtering and SSL/TLS Decryption Overview
 - Security Logs and Traffic Analysis
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Module 6: VPN Technologies

- VPN Fundamentals and Site-to-Site IPsec VPN
 - IKE Phase 1 and Phase 2, Encryption and Authentication Methods
 - Tunnel Interface and Route-Based VPN Implementation
 - IPsec VPN Monitoring and Troubleshooting
 - Global Protect Lab
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Module 7: Firewall Management & Troubleshooting

- High Availability (HA) Fundamentals and Active/Passive Configuration
 - Panorama Overview and Centralized Firewall Management
 - Palo Alto Packet Flow Architecture and Session Processing
 - Packet Capture and Troubleshooting Methodology
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Career Outcome

After completing this program, students will develop practical firewall administration, security policy implementation, and troubleshooting skills required for security-focused roles such as:

- Firewall Engineer (L1)
- Network Security Engineer
- Security Operations Engineer
- IT Infrastructure Security Engineer

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